Dauphin Island Nuclear Power Plant
Business Report

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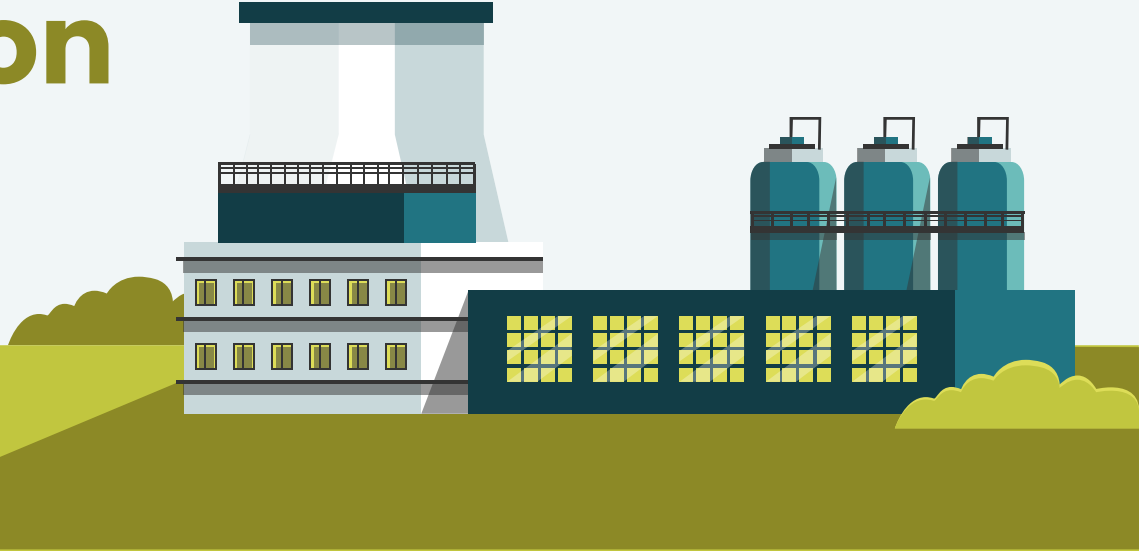
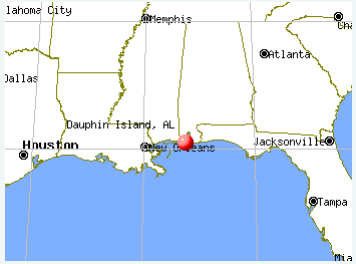
01

Background Information



Background Information

- Dauphin Island, Alabama
- Supplies power to Alabama and surrounding areas
- Nuclear Regulatory Commission has guidelines that must be followed
    - Reduce operations at 85°F water temp.
    - Prevent harm to environment around plant
- Assess likelihood of water temp. reaching the 85° threshold
- Strategies and recommendations



02

Methods

Data Collection

- Weather Stations
  - 2 stations on Dauphin Island
  - 1 station 20 miles to the SE in the Gulf of Mexico
- Data sources
  - NOAA (Through National Data Buoy Center)
  - NCEI (National Centers for Environmental Information)
- Downloaded all text files and added them to a new folder

- Data was public and came from reliable resources

Data Cleaning and Analysis

- Data merged into one data set
- Columns renamed, reshaped, and combined
  - Dplyr – used for cleaning
  - Lubridate – make date/time values easier to work with
  - Readr – read in the data sets
  - Ggplot2 – graph our data
- Summarize and maximum were the most used functions for analysis

Missing Data

	Initial	Kept	Removed
Number	288,915	236,000	52,915
Percentage	100%	81.68%	18.32%

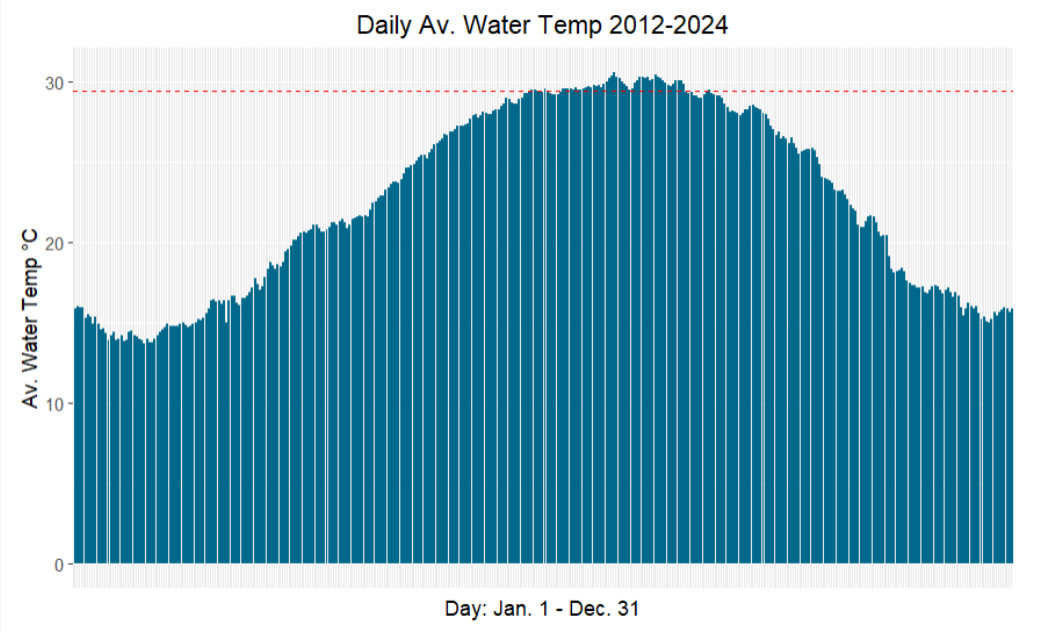
- 288,915 data points initially
- Incomplete/missing
  - Rows with 99/999 were removed
- Sufficient data for analysis – 20+ observations per day
- Must also take into account the different weather stations used – 2021 had the lowest number of “hot days” and a different station

03

Results

Daily Temperature Graph

- Shows the daily average water temperature from 2012-2024
- Months with values exceeding the maximum
  - June: 2
  - July: 24
  - August: 25
  - September: 2
- Would expect water temperature to reach maximum at least 50 times a year – especially in summer months



Confidence Intervals

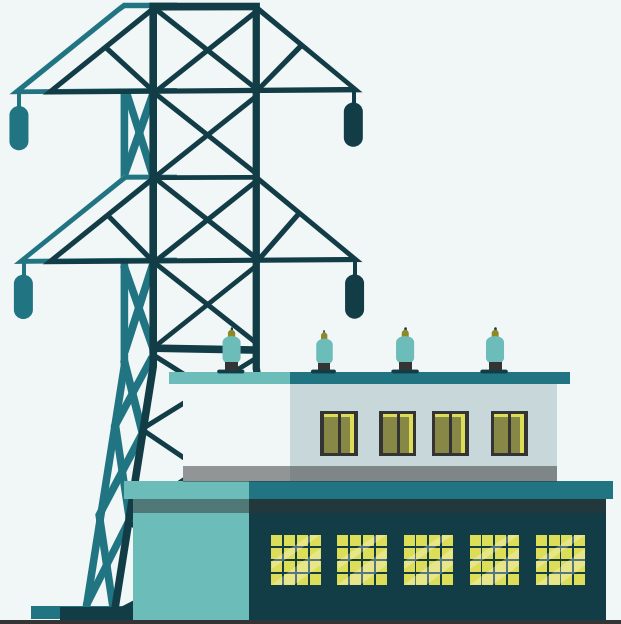
- The lm function was used to predict the number of days over the next five years that reach the 85° threshold
- This is a 95% confidence interval
- All intervals are greater than 0, so we would conclude that there would be a shutdown each of the next five years

Year	Predicted Number of Days	Lower Confidence Limit	Upper Confidence Limit
2025	99.3	40.8	157.8
2026	100.2	39.9	160.5
2027	101.1	38.9	163.3
2028	101.9	37.6	166.2
2029	102.8	36.2	169.3

Recommendations

- Recommend doing more research!!!
- Find new ways to cool water
  - Cooler water underground
  - Developing rapid sea water cooling plans
  - Building more cooling towers (water-free towers?)
- Compare to other plants around the world (France in particular)
- Research electricity use in Alabama and the surrounding areas
- Lobby for more climate change action and environmental protection

Thank you!



Do you have any questions?

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Resources

Bandyk, M. (2020, September 10). *For nuclear plants operating on thin margins, growing climate risks prompt tough choices*. Utility Dive. https://www.utilitydive.com/news/for-nuclear-plants-operating-on-thin-margins-growing-climate-risks-prompt/584883/

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